

Data Science for Economists

Lecture 6: Misc Base R

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Introduction

Agenda

Today we will cover a bunch of miscellaneous topics that don't quite fit anywhere else.

I will also mention some functions and tips and tricks that I find useful.

Pipeline

We're in (New) Pipe-land Now

Now, there is a pipe `▷` (`|>`) even in basic R (no package needed)! It lets you send (i.e. "pipe") intermediate output to another command.

In other words, it allows us to chain together a sequence of simple operations and thereby implement a more complex operation.

I want to reiterate how cool pipes are, and how using them can dramatically improve the experience of reading and writing code. Compare:

```
## These next two lines of code do exactly the same thing.
```

```
mpg ▷ subset(manufacturer="audi") ▷ aggregate(hwy ~ model, FUN = mean)  
aggregate(subset(mpg, manufacturer="audi"), hwy ~ model, FUN = mean)
```

The first line reads from left to right, exactly how I thought of the operations in my head.

- Take this object (`mpg`), do this (`subset`), then do this (`aggregate`), etc.

The second line totally inverts this logical order (the final operation comes first!)

- Who wants to read things inside out?

An aside on pipes: |> (cont.)

The piped version of the code is even more readable if we write it over several lines. Here it is again and, this time, I'll run it for good measure so you can see the output:

```
mpg |>
  subset(manufacturer=="audi") |>
  aggregate(hwy ~ model, FUN = mean)
```

```
##           model      hwy
## 1           a4 28.28571
## 2 a4 quattro 25.75000
## 3 a6 quattro 24.00000
```

Remember: Using vertical space costs nothing and makes for much more readable/writeable code than cramming things horizontally.

PS — The pipe is originally from the **magrittr** package, which can do some other cool things if you're inclined to explore. It's been so popular that even SQL is adopting a pipe syntax. SQL!

It is a very powerful tool. People used to use `%>%` in **magrittr**. However, it is so popular that the base R starts to have it as `|>` after R version 4.1.0 in 2021!

Dates and Times

Dates and Time

In order to work with dates and times correctly, they need to be specified as a date or time.

You will likely work with dates more often than times, so that is where more attention will be placed.

Understanding exactly how dates work requires understanding a bit more about the advanced aspects of R; however, to get the basics does not require this.

The default format for dates in R is `YYYY-MM-DD`.

```
# Convert to date with base function
```

```
as.Date("2021-09-14")
```

```
## [1] "2021-09-14"
```

```
Sys.Date()
```

```
## [1] "2025-10-21"
```

If you have a date not in the format listed above, you have to give the `as.Date()` function the "format" argument. [See this link for different formats.](#)

Lubridate

While we have been working in base `R` almost exclusively so far, I would highly recommend using the `lubridate` package.

Working with dates and times is much easier than in base `R`.

```
library(lubridate)
```

```
# Convert to date using lubridate functions  
ymd("2021-09-14")
```

```
## [1] "2021-09-14"
```

```
mdy("09/14/2021")
```

```
## [1] "2021-09-14"
```

```
today()
```

```
## [1] "2025-10-21"
```

Lubridate (Cont.)

```
# Extract a specific feature of the date  
year(today())
```

```
## [1] 2025
```

```
month(today())
```

```
## [1] 10
```

```
week(today())
```

```
## [1] 42
```

```
# More detailed date  
ymd_hms("2017-01-31 20:11:59")
```

```
## [1] "2017-01-31 20:11:59 UTC"
```

```
mdy_hm("01/31/2017 08:01")
```

```
## [1] "2017-01-31 08:01:00 UTC"
```

Date Comparison

It is easy to do comparison with dates. You can directly use logical comparisons `>` and `<`.

```
ymd("2021-09-14") > ymd("2021-08-12")
```

```
## [1] TRUE
```

```
ymd("2021-09-14") < "2021-08-12"
```

```
## [1] FALSE
```

```
# R can understand comparison between characters formatted as Date  
"2021-09-14" > "2021-08-12"
```

```
## [1] TRUE
```

```
"2021-09-14" < "2021-08-12"
```

```
## [1] FALSE
```

This is very handy if you need to do logical indexing with dates!

More About Dates and Times

There are a lot of little quirks when it comes to dates and times that usually you will not have to know. God bless you if you're ever working with time zones, spring forward, etc.

This is where the skill of Googling/LLM really comes in handy.

I would recommend also reading [Chapter 16](#) in *R for Data Science* that is listed on the syllabus.

Read/Write Files

Reading in Files

Because we've been working with simulated data or preloaded data, we have yet to have to read in data. There are four different types of files we will talk about today:

- csv: Work for all programs. Plain text (human-readable). Really close to Excel. Slow to write/read and large size.
- rds: Originated in R. There are special packages to load it into Python and Julia. Fast to write/read and small size.
- feather: Also works for Python and Julia. Very fast to write/read and medium size.
- fst: Originated in R. There are special packages to load it into Python and Julia. Ultra-fast to read/write even for massive datasets; small-medium (adjustable compression).

The most common function you will use is `read.csv()`.

- While you might want to read in Excel spreadsheets, I would highly recommend minimizing how much you use Excel for R.

To read in a file, you need to know where the file is located. You will need to set your "working directory."

```
# Get current working directory  
getwd()
```

```
## [1] "D:/R/ECON370/lec6"
```

```
# Set a new working directory  
setwd("D:/R/ECON370/lec6/")  
getwd()
```

```
## [1] "D:/R/ECON370/lec6"
```


See Files in Directory

To see the names of the files in a certain directory, use the `list.files()` functions.

```
# list all files/folders in the current working directory
```

```
list.files()
```

```
## [1] "06-miscR.html"          "06-miscR.pdf"
## [3] "06-miscR.Rmd"           "blp_data.csv"
## [5] "blp_data.rds"           "blp_data2.csv"
## [7] "blp_data2.rds"          "Lec6_2024"
## [9] "Lecture6Markdown_complete.Rmd" "Lecture6Markdown_template.Rmd"
## [11] "Lecture6Markdown_template_fill.Rmd" "Lecture6Script.R"
## [13] "libs"                   "OTC_data.feather"
## [15] "OTC_data.fst"
```

```
# list all files/folders in the specified working directory
```

```
# list.files("D:/R/ECON370")
```

```
list.files("D:/R/ECON370/HW1")
```

```
## [1] "2024"                  "Assignment1.pdf" "Assignment1.Rmd" "marsh"
```

Read in CSV

To read in the CSV listed above (about market share data and prices), use the function

`read.csv()`:

```
# OTC_data = read.csv("https://github.com/JiaxiLi1995/ECON370/raw/refs/heads/main/lec  
# write.csv(OTC_data, file = "blp_data.csv", row.names = F)  
OTC_data = read.csv("blp_data.csv")
```

We can save the csv using the function `write.csv()`:

```
write.csv(OTC_data, file = "blp_data2.csv", row.names = F)
```

A Few Things on CSV

CSV stands for "comma-separated values" i.e. the data values are separated by commas.

If you notice in the help documentation for `read.csv`, it assumes `sep=","`.

If this is not true, you might have to change the `sep` argument.

- e.g. Tab separated values are also common.

However, don't worry too much about this. While you need to understand it a little, we will be using `read_csv()` from the `tidyverse` and/or `fread()` from the `data.table` package, which are superior to `read.csv`.

rds

The rds files are native to R. We can try to save as rds file and read it with `saveRDS()` and `saveRDS()`:

```
# rds files
saveRDS(OTC_data, file = "blp_data.rds")
OTC_data_rds = readRDS("blp_data.rds")
```

You can also compress the file when saving it:

```
saveRDS(OTC_data_rds, file = "blp_data2.rds", compress = T)
```

feather

Feather files are designed for fast, efficient, and easy data interchange between languages.

`.feather` is usually the file extension. We will use a special package, `arrow`, to save (`write_feather()`) and load (`read_feather()`) feather file.

```
# feather files
```

```
library(arrow)
```

```
write_feather(OTC_data, "OTC_data.feather", compression = T)
```

```
OTC_data_feather = read_feather("OTC_data.feather")
```

fst

We can also work with `.fst` file in R. We will use `arrow` package to save (`write_fst()`) and load (`read_fst()`) `fst` file.

```
# fst files  
library(fst)  
write_fst(OTC_data, "OTC_data.fst")  
  
OTC_data_fst = read_fst("OTC_data.fst")
```

Speed Comparison

Unit: microseconds

##	expr	min	lq	mean	median	uq	max	neval
##	rds_read	629.400	692.802	737.5811	753.251	763.200	838.702	10
##	csv_read	5212.301	5437.301	5736.0307	5622.000	6080.900	6269.501	10
##	feather_read	4458.400	4600.901	5728.5109	5091.752	6080.601	10104.701	10
##	fst_read	447.401	486.801	539.3808	555.301	577.500	605.701	10

Unit: microseconds

##	expr	min	lq	mean	median	uq	max
##	rds_save	2568.601	2764.901	2865.5411	2827.4510	2927.101	3331.502
##	csv_save	18958.601	19596.701	21504.1412	22096.6010	22706.601	24244.602
##	feather_save	3147.701	3315.800	3949.3211	3752.8515	4566.801	5434.400
##	fst_save	623.301	681.701	852.6709	772.1005	1082.801	1196.900
##	neval						
##	10						
##	10						
##	10						
##	10						

Web Scraping

Internet as Data

1. Server-side

- Data stored at the server, which sends HTML code with data in it to us
- **Our process:** trudging through CSS selectors

2. Client-side

- Our browser requests data from the server, server sends specific info we asked for
- **Our process:** pingng an API endpoint

We will only demonstrate one example with `rvest`, which is server-side scrapping. There are more resources on webscrapping:

- Grant R. McDermott's lecture on webscrapping [section 1](#) and [section 2](#)
- [rvest](#)

We will use `rvest`, `dplyr` and `janitor` here (`lubridate` is already loaded):

```
library(rvest)
library(dplyr)
library(janitor)
```

Before we get started

1. Be a good internet user
2. It's easy to accidentally kill some poor website
3. It's probably legal?

Some website puts more limit to webscrapping (such as X, formally known as Twitter).

See the warnings from [the writer of `rvest`](#).

If you request a lot of data with scrapping, be polite with something like `Sys.sleep(2)` in between.

Use Rscript (`.R`) file to scrap data **only once** and save the data to some file (such as `.rds`). Then, read in the data file for your project. **Try not to scrap over and over again in your `.Rmd` file!**

HTML Basics

Here's some simple HTML:

```
<html>
<head>
  <title>Page title</title>
</head>
<body>
  <h1 id='first'>A heading</h1>
  <p>Some text &amp; <b>some bold text.</b></p>
  <img src='myimg.png' width='100' height='100'>
</body>
```

HTML Basics

Here's some simple HTML:

```
<html>
<head> ## head is an element #<<
  <title>Page title</title>
</head>
<body>
  <h1 id='first'>A heading</h1>
  <p>Some text &amp; <b>some bold text.</b></p>
  <img src='myimg.png' width='100' height='100'>
</body>
```

HTML Basics

Here's some simple HTML:

```
<html>
<head>
  <title>Page title</title>
</head>
<body> ## as is body #<<
  <h1 id='first'>A heading</h1>
  <p>Some text &amp; <b>some bold text.</b></p>
  <img src='myimg.png' width='100' height='100'>
</body>
```

HTML Basics

Here's some simple HTML:

```
<html>
<head>
  <title>Page title</title>
</head>
<body>
  <h1 id='first'>A heading</h1> ## id='first' is an attribute #<<
  <p>Some text &amp; <b>some bold text.</b></p>
  <img src='myimg.png' width='100' height='100'>
</body>
```

HTML Basics

Here's some simple HTML:

```
<html>
<head>
  <title>Page title</title>
</head>
<body>
  <h1 id='first'>A heading</h1>
  <p>Some text &amp; <b>some bold text.</b></p> ## stuff in between is contents #<<
  <img src='myimg.png' width='100' height='100'>
</body>
```

Lucky for us: we don't need to write HTML. Just read it.

HTML Basics - Tags

- Tags all start with `<tag>` and end with `</tag>`
- Every page is in an `<html>` element with 2 children:
 - `<head>` contains metadata
 - `<body>` contains content you see
- Block tags form the overall structure of a page
 - `<h1>` provides a heading
 - `<p>` is a paragraph
- Inline tags like `<b>` (bold), `<i>` (italics), and `<a>` (links) exist
- Just a sample of tags, can look up others you don't know
- The rest is the content

Example: Scraping Wikipedia

Let's imagine we want to scrape the [Men's 100 metres world record progression](https://en.wikipedia.org/wiki/Men%27s_100_metres_world_record_progression) page on Wikipedia.

In particular, we want to get the information from the 3 main tables on the webpage.

Here's what happens when we give R no instructions at all:

```
raw_wiki = read_html("https://en.wikipedia.org/wiki/Men%27s_100_metres_world_record_progression")
raw_wiki

## {html_document}
## <html class="client-nojs vector-feature-language-in-header-enabled vector-feature-language-in-current-page" ...
## [1] <head>\n<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" ...
## [2] <body class="skin--responsive skin-vector skin-vector-search-vue mediawiki" ...

class(raw_wiki)

## [1] "xml_document" "xml_node"
```

Parsing HTML: Inspecting Web Pages

We can use `inspect` to get a better sense of what is available on a given webpage. What tags can we use to grab the desired information from our wikipedia page?

It looks like we want to grab information from tables with the class "wikitable."

Brief Aside: CSS Selectors

We can parse HTML using **CSS Selectors**, which define patterns for locating HTML elements.

CSS Selectors are pretty complex, and we're going to keep it light today. If you want to learn more, check out the [CSS Diner](#). Additionally, if CSS Selectors are giving you tons of trouble, check out [SelectorGadget](#).

4 selectors to know:

1. `p` selects all `<p>` elements
2. `.title` selects all elements with `class` "title"
3. `p.special` selects all `<p>` elements with `class` "special"
4. `#title` selects the unique element with id attribute that equals "title"

Getting Our Tables

We can use the selector `table` to select all `<table>` elements:

```
raw_wiki > html_elements("table")
```

```
## {xml_nodeset (15)}
## [1] <table class="box-Unreferenced_section plainlinks metadata ambox ambox-c ...
## [2] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Athlete\n</ ...
## [3] <table class="wikitable" style="text-align: left;"><tbody>\n<tr>\n<td st ...
## [4] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Wind\n</th> ...
## [5] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Wind\n</th> ...
## [6] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Athlete\n</ ...
## [7] <table class="nowraplinks mw-collapsible autocollapse navbox-inner" styl ...
## [8] <table class="nowraplinks navbox-subgroup" style="border-spacing:0"><tbo ...
## [9] <table class="nowraplinks hlist mw-collapsible autocollapse navbox-inner ...
## [10] <table class="nowraplinks navbox-subgroup" style="border-spacing:0"><tbo ...
## [11] <table class="nowraplinks navbox-subgroup" style="border-spacing:0"><tbo ...
## [12] <table class="nowraplinks mw-collapsible uncollapsed navbox-inner" style ...
## [13] <table class="nowraplinks navbox-subgroup" style="border-spacing:0"><tbo ...
## [14] <table class="nowraplinks navbox-subgroup" style="border-spacing:0"><tbo ...
## [15] <table class="nowraplinks navbox-subgroup" style="border-spacing:0"><tbo ...
```

Alas, so many other things.

Trying again

We want tables with class wikitable: `table.wikitable` should work!

```
raw_wiki ▷ html_elements("table.wikitable")
```

```
## {xml_nodeset (5)}  
## [1] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Athlete\n</t ...  
## [2] <table class="wikitable" style="text-align: left;"><tbody>\n<tr>\n<td sty ...  
## [3] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Wind\n</th>\n ...  
## [4] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Wind\n</th>\n ...  
## [5] <table class="wikitable"><tbody>\n<tr>\n<th>Time\n</th>\n<th>Athlete\n</t ...
```

Looks better!

Accessing the actual info

We still don't have the table info. `html_table()` can help:

```
table_dfs = raw_wiki %>% html_elements("table.wikitable") %>%  
  html_table()  
table_dfs[[1]]
```

```
## # A tibble: 21 × 5  
##   Time Athlete      Nationality `Location of races`   Date  
##   <dbl> <chr>      <chr>      <chr>      <chr>  
## 1  10.8 Luther Cary    United States  Paris, France    July 4, 1...  
## 2  10.8 Cecil Lee      United Kingdom Brussels, Belgium September...  
## 3  10.8 Étienne De Ré   Belgium       Brussels, Belgium August 4,...  
## 4  10.8 L. Atcherley    United Kingdom Frankfurt/Main, Germany April 13,...  
## 5  10.8 Harry Beaton    United Kingdom Rotterdam, Netherlands August 28...  
## 6  10.8 Harald Anderson-Arbin Sweden         Helsingborg, Sweden August 9,...  
## 7  10.8 Isaac Westergren Sweden         Gävle, Sweden    September...  
## 8  10.8 Isaac Westergren Sweden         Gävle, Sweden    September...  
## 9  10.8 Frank Jarvis    United States  Paris, France    July 14, ...  
## 10 10.8 Walter Tewksbury United States  Paris, France    July 14, ...  
## # i 11 more rows
```

...this is crazy! `R` is cool sometimes.

General Workflow

Your workflow using `rvest`: get html --> get desired html elements --> break into individual elements --> turn into table/text/numbers/whatever.

Useful commands for getting individual elements:

- `html_text2()` gets plain text contents of an HTML element
- `html_attr()` gets attributes from an HTML element (e.g., links)
- `html_table()` creates a data.frame from a table in an HTML element

A Little Cleanup

Let's get our data frames in working order here:

```
table_dfs_int = raw_wiki > html_elements("table.wikitable") >  
  html_table()
```

```
table_dfs = lapply(table_dfs_int[c(1,3,4)], # drop unwanted tables  
  function(x) x >
```

```
    clean_names() > ## fix colnames, from the janitor package  
    mutate(date = mdy(date))) ## from lubridate
```

```
table_dfs[[1]]
```

```
## # A tibble: 21 × 5
```

##	time	athlete	nationality	location_of_races	date
##	<dbl>	<chr>	<chr>	<chr>	<date>
## 1	10.8	Luther Cary	United States	Paris, France	1891-07-04
## 2	10.8	Cecil Lee	United Kingdom	Brussels, Belgium	1892-09-25
## 3	10.8	Étienne De Ré	Belgium	Brussels, Belgium	1893-08-04
## 4	10.8	L. Atcherley	United Kingdom	Frankfurt/Main, Germany	1895-04-13
## 5	10.8	Harry Beaton	United Kingdom	Rotterdam, Netherlands	1895-08-28
## 6	10.8	Harald Anderson-Arbin	Sweden	Helsingborg, Sweden	1896-08-09
## 7	10.8	Isaac Westergren	Sweden	Gävle, Sweden	1898-09-11
## 8	10.8	Isaac Westergren	Sweden	Gävle, Sweden	1899-09-10
## 9	10.8	Frank Jarvis	United States	Paris, France	1900-07-14
## 10	10.8	Walter Tewksbury	United States	Paris, France	1900-07-14

Combined Data

```
wr100 = rbind(  
  table_dfs[[1]] > select(time, athlete, nationality, date) >  
    mutate(era="Pre-IAAF"),  
  table_dfs[[2]] > select(time, athlete, nationality, date) >  
    mutate(era="Pre-automatic"),  
  table_dfs[[3]] > select(time, athlete, nationality, date) >  
    mutate(era="Modern")  
)  
head(wr100)
```

```
## # A tibble: 6 × 5
```

##	time	athlete	nationality	date	era
##	<dbl>	<chr>	<chr>	<date>	<chr>
## 1	10.8	Luther Cary	United States	1891-07-04	Pre-IAAF
## 2	10.8	Cecil Lee	United Kingdom	1892-09-25	Pre-IAAF
## 3	10.8	Étienne De Ré	Belgium	1893-08-04	Pre-IAAF
## 4	10.8	L. Atcherley	United Kingdom	1895-04-13	Pre-IAAF
## 5	10.8	Harry Beaton	United Kingdom	1895-08-28	Pre-IAAF
## 6	10.8	Harald Anderson-Arbin	Sweden	1896-08-09	Pre-IAAF

Presidential Speech and Fed

There are many textual data online are publicly available.

If you are interested, you can scrap US presidential speech from The American Presidency Project (which is the most complete archive) using `rvest`.

You can enter "Webscrapping presidential speech from the American Presidency Project (which is the most complete archive) using rvest (R)" in ChatGPT to get the structured code.

You can also scrap the Federal Reserve Annoucements.

Regular Expressions

Parsing Text

Parsing text to find patterns or extract data is hard.

There are a group of functions that can be used for this: the `grep()` functions.

`grep()` takes a pattern and character vector and returns a vector of indexes for which elements contained the pattern.

`grepl()` takes a pattern and character vector and returns a logical vector for if each element contains the pattern.

`sub()` takes a pattern, a replacement, and a character vector and returns the character vector with the first occurrence of the pattern replaced with the replacement.

`gsub()` is exactly like `sub()` except that it replaces all occurrences.

Parsing Text: Examples

```
statecountry_names = c('Florida', 'Germany', 'Georgia', 'Geniva',  
                        'Istanbul', 'NewZealand', 'Australia')  
grep("G", statecountry_names)
```

```
## [1] 2 3 4
```

```
grepl("G", statecountry_names)
```

```
## [1] FALSE TRUE TRUE TRUE FALSE FALSE FALSE
```

```
grepl("A", statecountry_names, ignore.case = F)
```

```
## [1] FALSE FALSE FALSE FALSE FALSE FALSE TRUE
```

```
grepl("A", statecountry_names, ignore.case = T)
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

Parsing Text: Examples

```
sub("G", "A", statecountry_names)
```

```
## [1] "Florida"      "Aermany"      "Aeorgia"      "Aeniva"       "Istanbul"  
## [6] "NewZealand"   "Australia"
```

```
sub("G", "A", statecountry_names, ignore.case = T)
```

```
## [1] "Florida"      "Aermany"      "Aeorgia"      "Aeniva"       "Istanbul"  
## [6] "NewZealand"   "Australia"
```

```
gsub("G", "A", statecountry_names, ignore.case = T)
```

```
## [1] "Florida"      "Aermany"      "AeorAia"      "Aeniva"       "Istanbul"  
## [6] "NewZealand"   "Australia"
```

Regular Expressions

Sometimes there are patterns that you would like to detect in text that are more complicated than just a short character expression that you can program.

Or you'd like to do all combinations of something.

This is where regular expressions can be used.

Truthfully, we could spend an entire class just learning regular expressions.

As such, I am only going to briefly touch them and encourage you to look into them more on your own.

The plus side is they are pretty universal across languages. That is, whether you're using `R` or `Python`, regular expressions still work the same way!

Regular Expressions: grepl Examples

Let's say we want to flag character strings that start with "The". The regular expression would be `^The`

```
test_char = c("The beginning.", "The end.", "The middle.", "Other.")
grepl("^The", test_char)
```

```
## [1] TRUE TRUE TRUE FALSE
```

Now, let's say we want to flag character strings that end with "end." The regular expression would be `end.$`

```
grepl("end.$", test_char)
```

```
## [1] FALSE TRUE FALSE FALSE
```

Find all vowels

```
grepl("[aeiou]", letters)
```

```
## [1] TRUE FALSE FALSE FALSE TRUE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [13] FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [25] FALSE FALSE
```


Regular Expressions: More grepl

Let's say we have a couple strings to match:

```
test_char = c("The beginning.", "The end.", "The middle.", "Other.")
grepl("beginning|end", test_char)
```

```
## [1] TRUE TRUE FALSE FALSE
```

We can also grab numbers from a string:

```
grepl("[0-9]", c(1:5, "I had 2 coffees today", "Hello World!"))
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE FALSE
```

Regular Expressions: Only csv

We have talked about how to use `list.files()` to get files names in a folder:

```
# list all files/folders in the specified working directory
```

```
list.files("D:/R/ECON370/lec6")
```

```
## [1] "06-miscR.html"          "06-miscR.pdf"
## [3] "06-miscR.Rmd"           "blp_data.csv"
## [5] "blp_data.rds"           "blp_data2.csv"
## [7] "blp_data2.rds"          "Lec6_2024"
## [9] "Lecture6Markdown_complete.Rmd" "Lecture6Markdown_template.Rmd"
## [11] "Lecture6Markdown_template_fill.Rmd" "Lecture6Script.R"
## [13] "libs"                   "OTC_data.feather"
## [15] "OTC_data.fst"
```

Suppose that I only want to have all the files with `.csv` extension (could be multiple files), we can use regular expressions:

```
list.files(
  path = "D:/R/ECON370/lec6",
  pattern = ".*\\.csv$",
  full.names = TRUE
)
```

Regular Expressions: More

Validating that a user supplied an email to a website form is famously a difficult problem. Here's one regex that a user on [StackOverflow](#) put forth:

```
pattern <- (?:[a-z0-9!#$%&'*/=?^_`{|}~-]+(?:[a-z0-9!#$%&'*/=?^_`{|}~-]+)|'(?:[\x01-  
  \x08\x0b\x0c\x0e-\x1f\x21\x23-\x5b\x5d-\x7f]|[\x01-\x09\x0b\x0c\x0e-\x7f]))'@(?:\a-z0-  
  9?)+a-z0-9?|((?:\a-z0-9|[0-9-~]|[\x01-\x08\x0b\x0c\x0e-\x1f\x21-\x5a\x53-\x7f]|[\x01-  
  \x09\x0b\x0c\x0e-\x7f]))+))
```

Regular Expressions: More

Regular expressions are very powerful for processing text data. Processing text data is a skill set in-and-of itself.

I use LLMs write my regexs and it is usually really accurate.

Lots of learning when it comes to programming is simply being aware of something so that you can look into it more when you actually need to use it.

If you want to learn more about regular expressions, [see the following cheat-sheet](#).

Functions That I Use

Misc Functions

We have seen:

- `rep()`: repeats a vector N times
 - Used a lot for preallocating vectors
- `sum()`: sums up vectors
- `seq()`: creates sequences
- `cbind()`: combine matrices or data.frames by columns
- `rbind()`: combine matrices or data.frames by rows
- `ncol()`: returns number of columns of data.frame or matrix
- `nrow()`: returns number of rows of data.frame or matrix
- `dim()`: returns dimensions of an array-like object

New Ones:

- `unique()`: takes the unique values of a vector
- `sort()`: sorts a vector
- `order()`: like sort, but gives the IDs rather than a sorted vector
- `substr()`: returns a string subsetting by the indexes supplied
- `strsplit()`: splits a string by a pattern
- `na.omit()`: remove the rows/observations containing NAs

Examples

```
rep(0,5)
```

```
## [1] 0 0 0 0 0
```

```
rep(1:10,2)
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10
```

```
rep(1:10,each=2)
```

```
## [1] 1 1 2 2 3 3 4 4 5 5 6 6 7 7 8 8 9 9 10 10
```

```
letter_samp = sample(letters,12,replace=T)  
letter_samp
```

```
## [1] "n" "y" "w" "c" "h" "p" "l" "y" "n" "c" "n" "g"
```

```
unique(letter_samp)
```

```
## [1] "n" "y" "w" "c" "h" "p" "l" "g"
```

Examples (Cont.)

```
norm_draws = rnorm(10)
norm_draws
```

```
## [1] -0.7729782  2.1499193 -1.3343536  0.4958705  1.2339762  0.6343621
## [7]  0.4120223  0.7935853 -0.1524106 -0.2288958
```

```
sort(norm_draws)
```

```
## [1] -1.3343536 -0.7729782 -0.2288958 -0.1524106  0.4120223  0.4958705
## [7]  0.6343621  0.7935853  1.2339762  2.1499193
```

```
order(norm_draws)
```

```
## [1]  3  1 10  9  7  4  6  8  5  2
```

```
sum(norm_draws)
```

```
## [1] 3.231097
```

```
seq(1,2.4,by=0.1)
```

```
## [1] 1.0 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8 1.9 2.0 2.1 2.2 2.3 2.4
```


Examples (Cont.)

```
rand_mat1 = matrix(rnorm(2*3),nrow=3)
rand_mat2 = matrix(rnorm(1*3),nrow=3)
rand_mat1
```

```
##           [,1]      [,2]
## [1,] -0.9007918  0.619283535
## [2,] -0.7350262 -0.006198262
## [3,] -1.4276858 -0.685706846
```

```
rand_mat2
```

```
##           [,1]
## [1,] -0.2793335
## [2,] -0.7827303
## [3,] -0.7789972
```

```
cbind(rand_mat1,rand_mat2)
```

```
##           [,1]      [,2]      [,3]
## [1,] -0.9007918  0.619283535 -0.2793335
## [2,] -0.7350262 -0.006198262 -0.7827303
## [3,] -1.4276858 -0.685706846 -0.7789972
```

Examples (Cont.)

```
rand_mat1 = matrix(rand_mat1,nrow=2)
rand_mat2 = matrix(rand_mat2,nrow=1)
rand_mat1
```

```
##           [,1]      [,2]      [,3]
## [1,] -0.9007918 -1.4276858 -0.006198262
## [2,] -0.7350262  0.6192835 -0.685706846
```

```
rand_mat2
```

```
##           [,1]      [,2]      [,3]
## [1,] -0.2793335 -0.7827303 -0.7789972
```

```
rbind(rand_mat1,rand_mat2)
```

```
##           [,1]      [,2]      [,3]
## [1,] -0.9007918 -1.4276858 -0.006198262
## [2,] -0.7350262  0.6192835 -0.685706846
## [3,] -0.2793335 -0.7827303 -0.778997240
```

Examples (Cont.)

```
ncol(rand_mat2)
```

```
## [1] 3
```

```
nrow(rand_mat2)
```

```
## [1] 1
```

```
dim(rand_mat2)
```

```
## [1] 1 3
```

```
my_string = "Hello, This is Econ 370, and my name is Jiaxi"  
substr(my_string,1,10)
```

```
## [1] "Hello, Thi"
```

```
strsplit(my_string,",")
```

```
## [[1]]
```

```
## [1] "Hello"           " This is Econ 370"      " and my name is Jiaxi"
```

Examples

```
# Create a data frame with some missing values
```

```
df = data.frame(  
  x = c(1, 2, NA, 4, 5),  
  y = c(NA, 10, 15, 20, NA)  
)
```

```
# remove all the NAs
```

```
na.omit(df)
```

```
##      x  y
```

```
## 2 2 10
```

```
## 4 4 20
```

Next lecture: Tidyverse
